

Lab 5: Hypothesis testing

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Basel temperature data

We read the data

```
temperature.basel<-read.table("basel-temperatures.txt",header=TRUE)
```

Two-side alternative

```
y<-temperature.basel$degrees  
t.test(y,mu=2,alternative = "two.sided")
```

```
##  
## One Sample t-test  
##  
## data: y  
## t = -2.0924, df = 109, p-value = 0.03873  
## alternative hypothesis: true mean is not equal to 2  
## 95 percent confidence interval:  
## 1.355288 1.982530  
## sample estimates:  
## mean of x  
## 1.668909
```

One-side alternative

```
t.test(y,mu=2,alternative = "less")
```

```
##  
## One Sample t-test  
##  
## data: y  
## t = -2.0924, df = 109, p-value = 0.01936  
## alternative hypothesis: true mean is less than 2  
## 95 percent confidence interval:  
## -Inf 1.931417  
## sample estimates:  
## mean of x  
## 1.668909
```

Saplings in a Finnish forest

We need to install the package `spatstat` for getting the data.

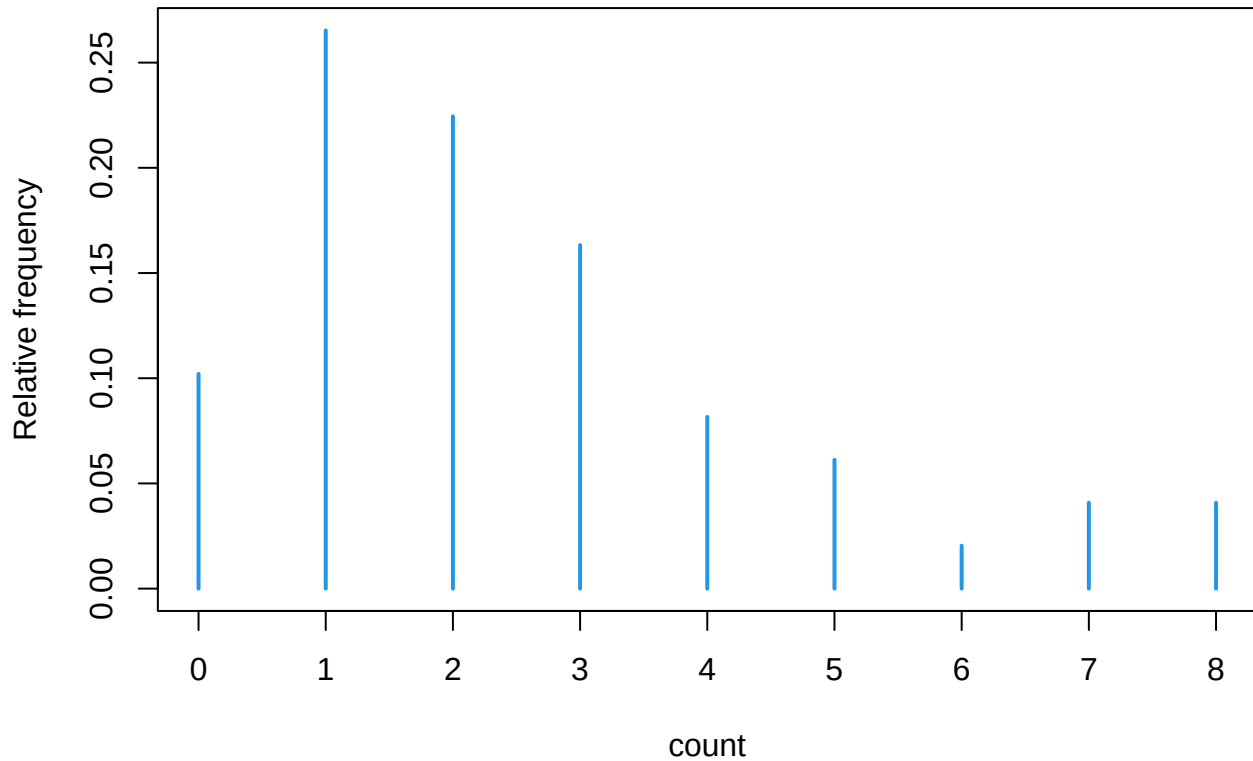
The dataset `finpines` record the locations of 126 pine saplings in a Finnish forest, their heights and their diameters.

Quadrat counts

```
quadrat<-read.table("quadratcounts.txt",header = TRUE)
y<-quadrat$counts
```

Empirical distribution

```
n<-length(y)
plot(table(y)/n,ylab="Relative frequency",xlab="count",col=4)
```



Is it the mean of the number of trees in a quadrat equal to 3 ?

We start by fitting a Poisson distribution

```
library(fitdistrplus)
```

```
## Loading required package: MASS
```

```
## Loading required package: survival
```

```
fit<-fitdistr(y,distr = "pois")
fit
```

```
## Fitting of the distribution ' pois ' by maximum likelihood
```

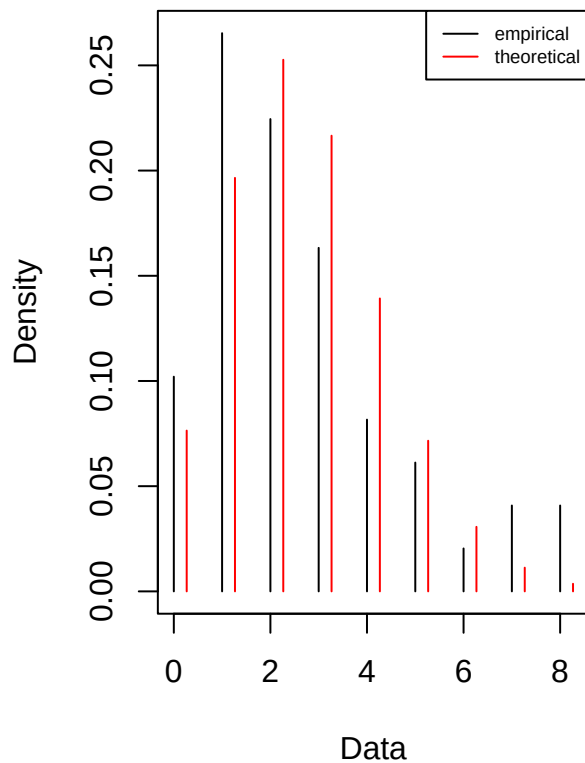
```
## Parameters:
```

```
##      estimate Std. Error
```

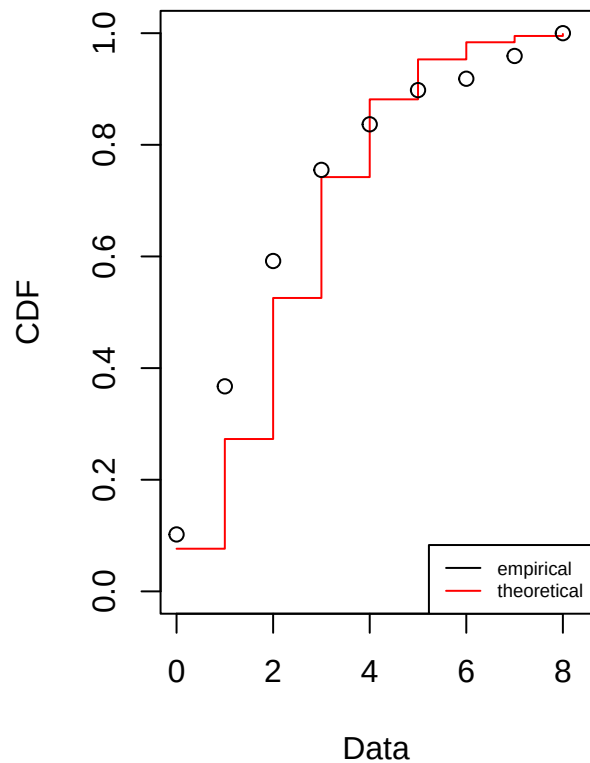
```
## lambda 2.571429  0.229081
```

```
plot(fit)
```

Emp. and theo. distr.



Emp. and theo. CDFs



Test by z-statistics and p-value

```
lambda.0<-3
lambda.hat<-fit$estimate[1]
se.lambda.hat<-fit$sd[1]
z<-(lambda.hat-lambda.0)/se.lambda.hat
poss<-pnorm(z)
```

P-value is 0.0306844. Conclusion ?

Geneva rainfall data

We read the data

```
rainfall<-read.table("geneva-rainfall.txt",header=TRUE)
```

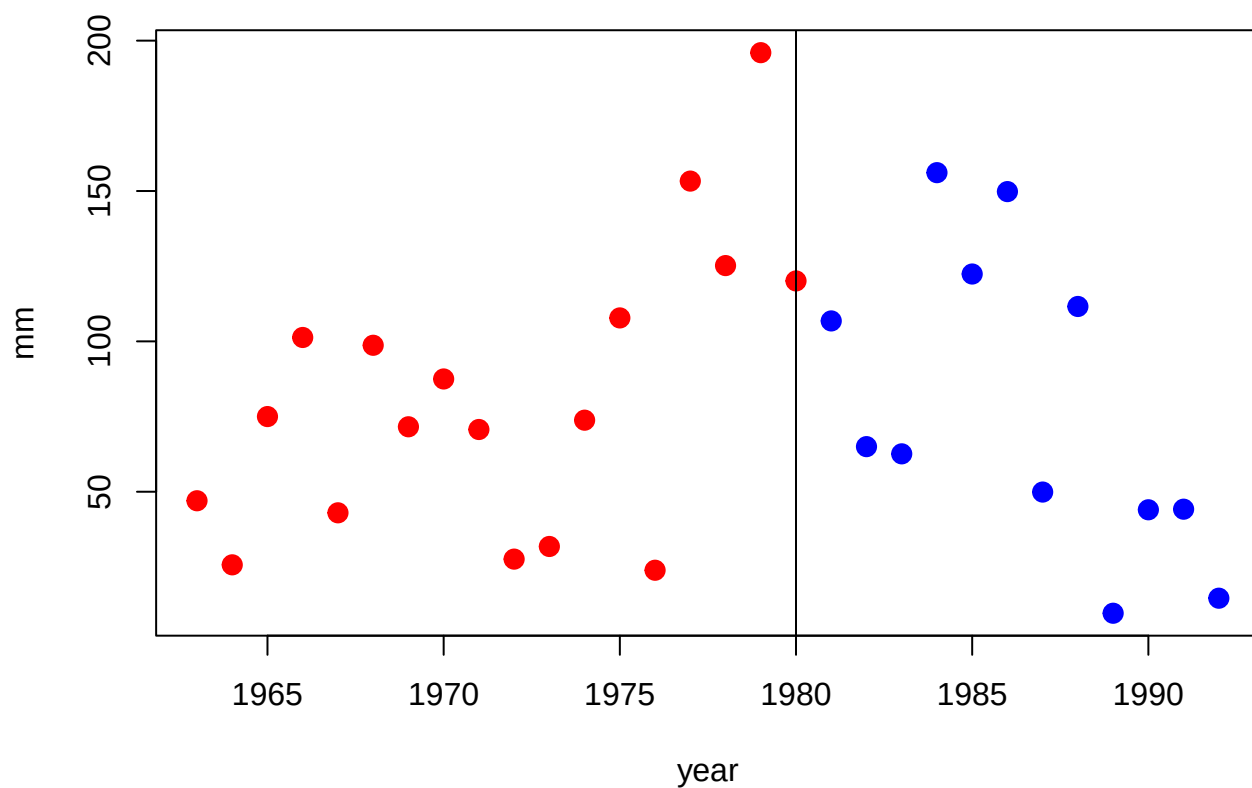
We define the two periods

```
per1 <- (rainfall[, "year"] <= 1980)
per2 <- !per1
```

Plot the original time series. Is there visual evidence of a shift?

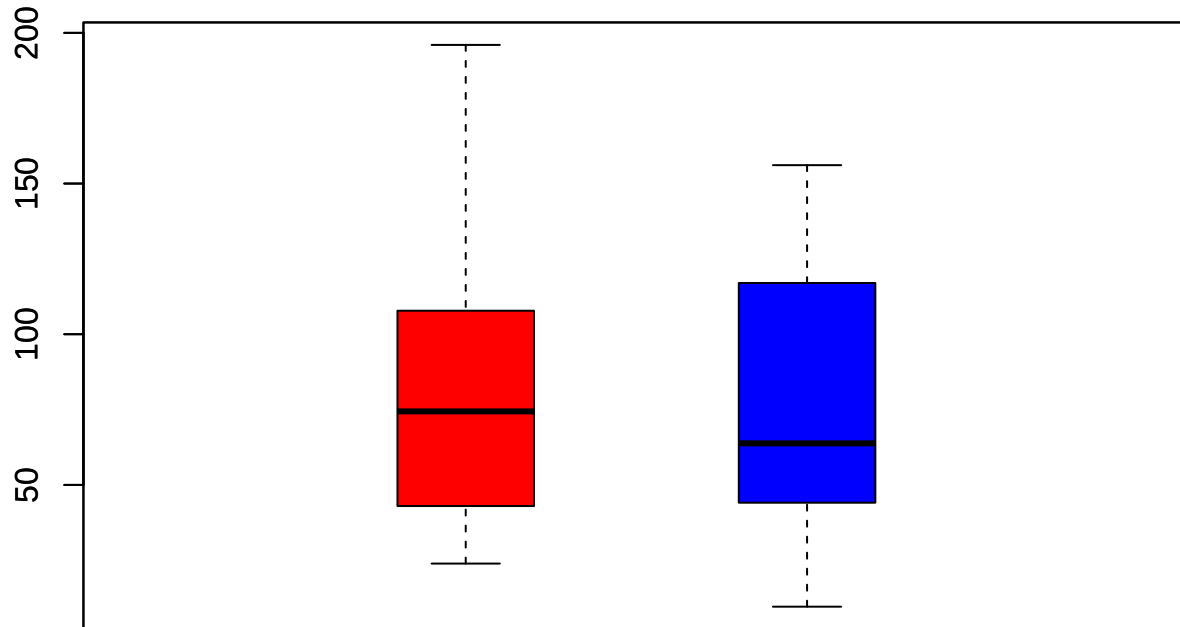
```
plot(x=rainfall[per1, "year"], y=rainfall[per1, "Geneva"], xlim=range(rainfall[, "year"]), ylim=range(rainfall[, "Geneva"]),
     xlab="year", ylab="mm")
points(x=rainfall[per2, "year"], y=rainfall[per2, "Geneva"], col="blue", pch=20, cex=2)
abline(v=1980)
title("Original Measurements")
```

Original Measurements



```
ylim<-range(rainfall[, "Geneva"])
x<-rainfall[per1, "Geneva"]
y<-rainfall[per2, "Geneva"]
boxplot(x, main="Rainfall", xlim=c(0,3), ylim=ylim, col="red")
boxplot(y, add = TRUE, at = 2, ylim=ylim, col="blue")
```

Rainfall



Welch's two-sided test

```
tt<-t.test(x=x,y=y,alternative="two.sided")
tt

##
##  Welch Two Sample t-test
##
## data:  x and y
## t = 0.22927, df = 22.792, p-value = 0.8207
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -33.49113  41.83557
## sample estimates:
## mean of x mean of y
##  82.22222  78.05000
```

Pooled variance two-sided test

```
tt.pooled<-t.test(x=x,y=y,alternative="two.sided",var.equal=TRUE)
tt.pooled

##
##  Two Sample t-test
##
## data:  x and y
## t = 0.23192, df = 28, p-value = 0.8183
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -32.67799  41.02244
## sample estimates:
## mean of x mean of y
##  82.22222  78.05000
```

Wilcoxon-Mann-Whitney test

```
tt.w<-wilcox.test(x=x,y=y,alternative="two.sided")
tt.w
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  x and y
## W = 113, p-value = 0.8513
## alternative hypothesis: true location shift is not equal to 0
```

Jura data

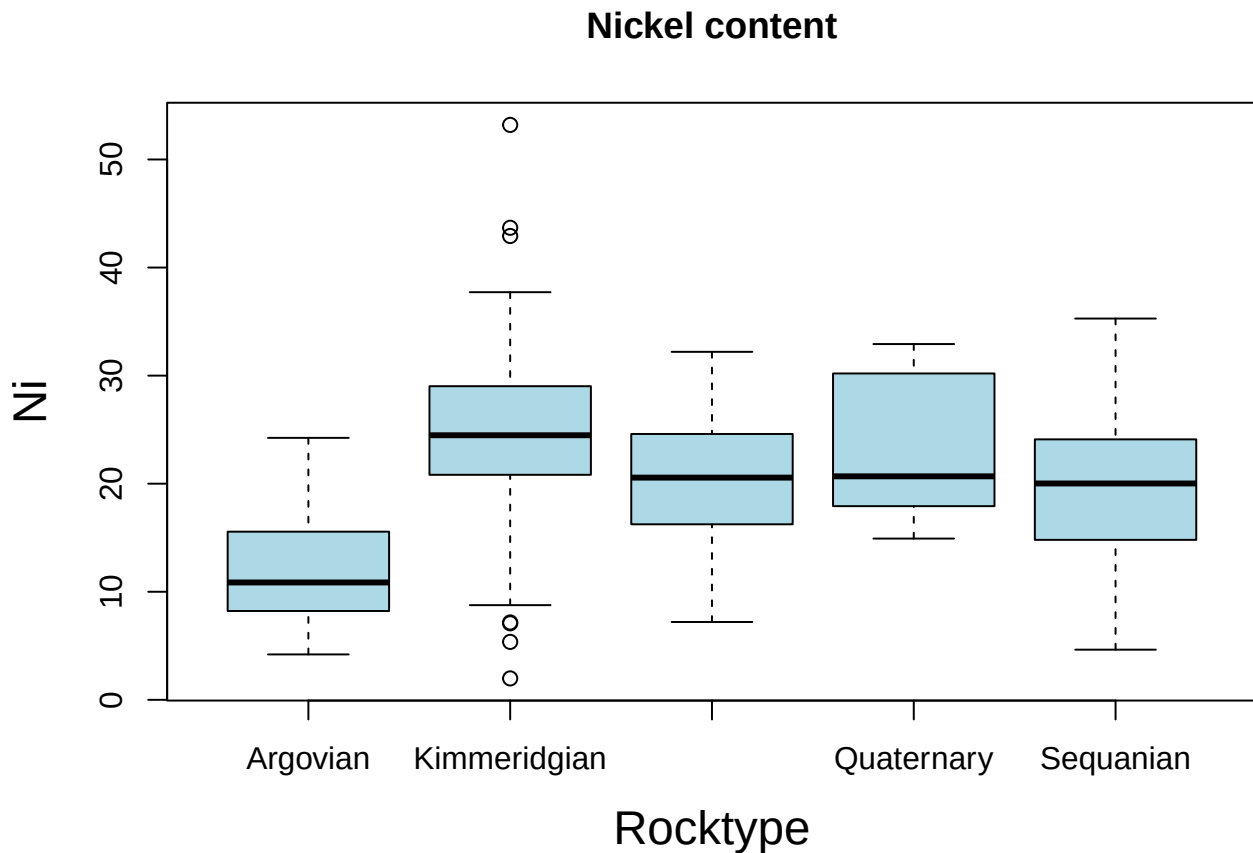
We read the data

```
jura<-read.csv("juraset.csv",dec = ",")
```

We transform the string variables into factor (categorical) variables

```
Rocktype<-as.factor(jura$Rock)
levels(Rocktype)<-c("Argovian", "Kimmeridgian", "Portlandian", "Quaternary", "Sequanian")
LandUse<-as.factor(jura$Land)
levels(LandUse)<-c("Forest", "Meadow", "Pasture", "Tillage")
```

```
attach(jura)
plot(Ni~Rocktype,main="Nickel content",cex.lab=1.5,col="lightblue")
```



Calculate the mean of each group

```
aggregate(Ni ~ Rocktype, FUN = mean)
```

```
##      Rocktype      Ni
## 1   Argovian 12.28105
## 2 Kimmeridgian 24.95984
## 3 Portlandian 20.41888
## 4 Quaternary 22.88667
## 5 Sequanian 18.80875
```

Calculate the standard deviation of each group

```
aggregate(Ni ~ Rocktype, FUN = sd)
```

```
##      Rocktype      Ni
## 1   Argovian 5.564566
## 2 Kimmeridgian 7.388552
## 3 Portlandian 5.660068
## 4 Quaternary 7.103325
## 5 Sequanian 7.566538
```

Anova table

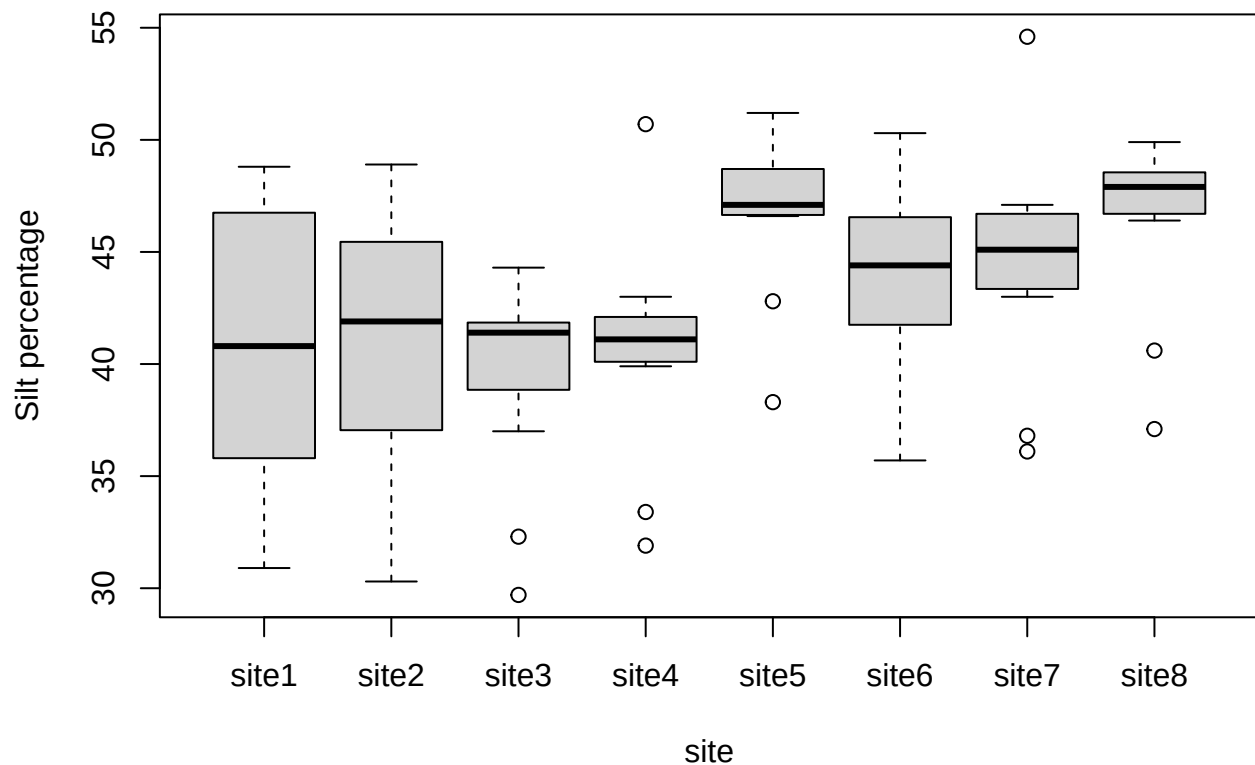
```
summary(aov(Ni~Rocktype))
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Rocktype      4   7735   1933.7    43.56 <2e-16 ***
## Residuals    354   15715     44.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Albudeite data

```
Albudeite<-read.table("silt.txt",header=FALSE)
```

```
names(Albudeite)<-c("silt","site")
attach(Albudeite)
boxplot(silt~site,ylab="Silt percentage")
```



```
summary(aov(silt~site))
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## site         7   600.1    85.73   3.432 0.00293 **
## Residuals    80 1998.4    24.98
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

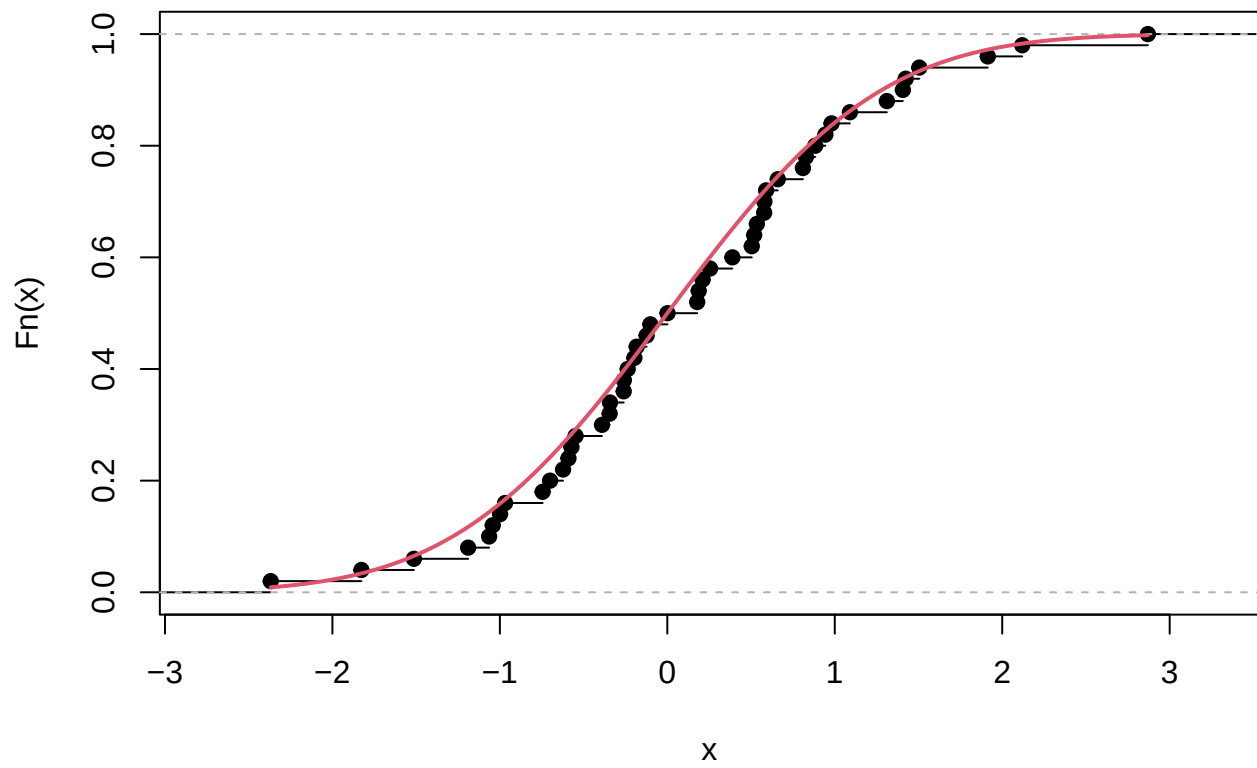
```
kruskal.test(silt~site)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  silt by site
## Kruskal-Wallis chi-squared = 22.467, df = 7, p-value = 0.002109
```

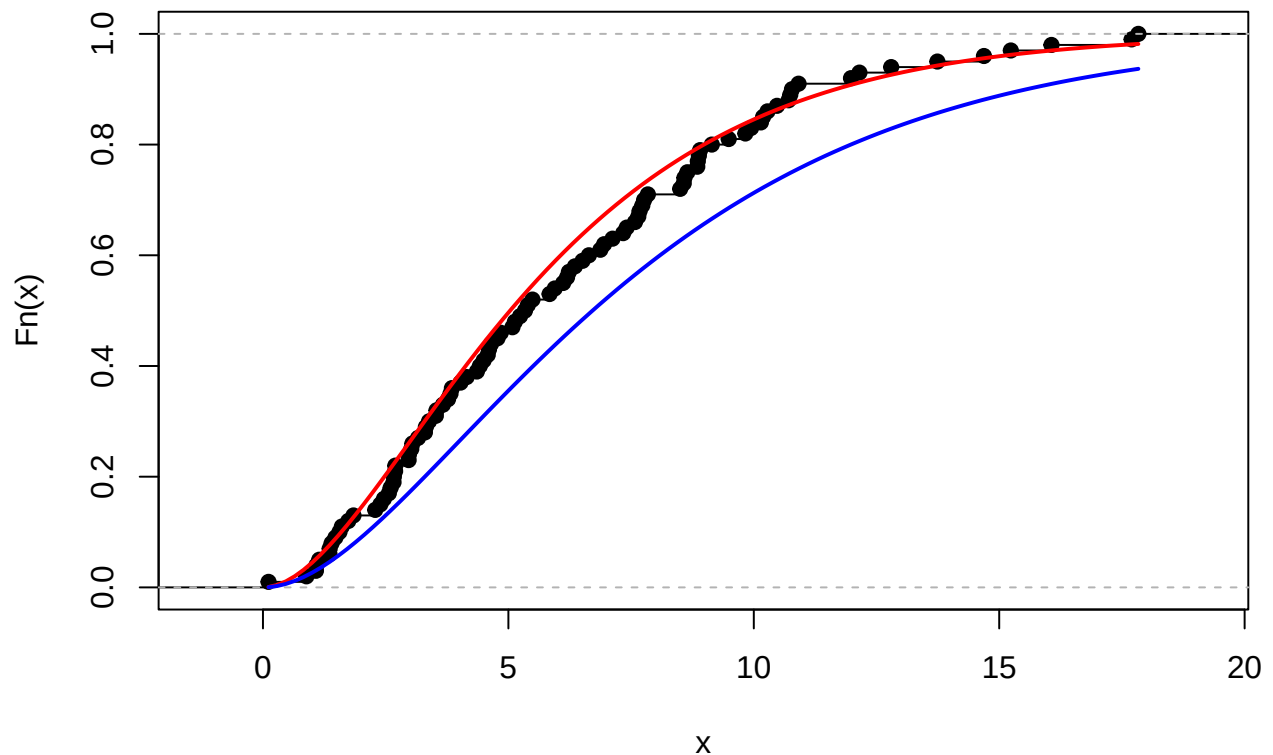
Goodnes-of-fit tests

Simulated example

```
set.seed(19)
y<-rnorm(50)
plot(ecdf(y),main="",verticals=FALSE)
curve(pnorm(x),from=min(y),to=max(y),add=TRUE,lwd =2,col=2)
```

```
set.seed(23)
y<-rgamma(100,shape =2, scale=3)
plot(ecdf(y),main="",verticals=FALSE)
curve(pgamma(x,shape =2, scale=3),
      from=min(y),to=max(y),add=TRUE,lwd =2,col="red")
curve(pgamma(x,shape =2, scale=4),
      from=min(y),to=max(y),add=TRUE,lwd =2,col="blue")
```



Kolmogorov-Smirnov test

```
ks.test(y, "pgamma", shape=2, scale=3)
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: y
## D = 0.071506, p-value = 0.686
## alternative hypothesis: two-sided
```

```
ks.test(y, "pgamma", shape=2, scale=4)
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: y
## D = 0.15362, p-value = 0.01783
## alternative hypothesis: two-sided
```

Anderson-Darling test

We install the required package

```
install.packages("goftest")
```

```
library(goftest)
ad.test(y, "pgamma", shape=2, scale=3)
```

```
##
## Anderson-Darling test of goodness-of-fit
```

```
## Null hypothesis: Gamma distribution
## with parameters shape = 2, scale = 3
## Parameters assumed to be fixed
##
## data: y
## An = 0.6227, p-value = 0.6264
```

```
ad.test(y, "pgamma", shape=2, scale=4)
```

```
##
## Anderson-Darling test of goodness-of-fit
## Null hypothesis: Gamma distribution
## with parameters shape = 2, scale = 4
## Parameters assumed to be fixed
##
## data: y
## An = 4.7016, p-value = 0.004008
```

Real data example: rainfall in Basel

```
rainfall.basel<-read.table("basel-rainfall.txt",header=TRUE)
```

We fit two distributions

```
library(fitdistrplus)
y<-rainfall.basel$mm
fit.gauss<-fitdist(y,distr = "norm")
fit.gamma<-fitdist(y,distr = "gamma")
```

```
gofstat(list(fit.gauss,fit.gamma))
```

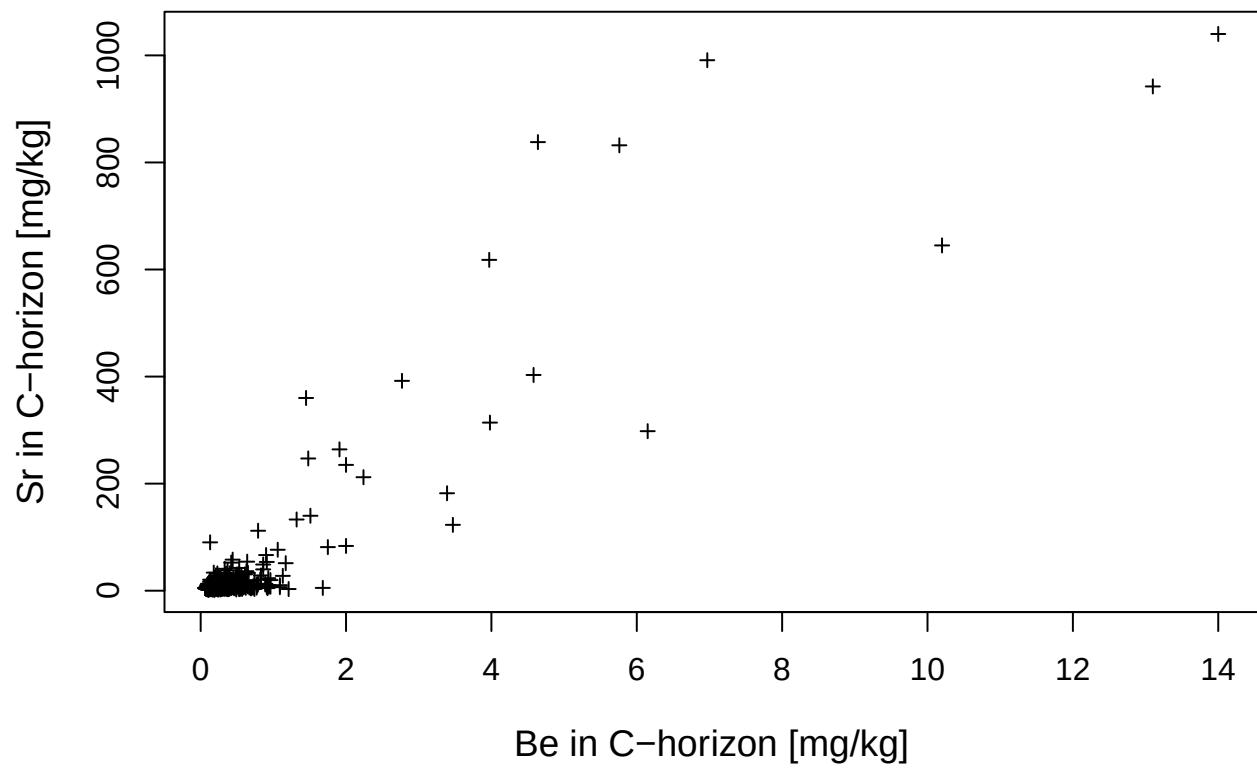
```
## Goodness-of-fit statistics
##                               1-mle-norm 2-mle-gamma
## Kolmogorov-Smirnov statistic 0.09543448 0.07231980
## Cramer-von Mises statistic   0.18909204 0.05918861
## Anderson-Darling statistic   1.19416447 0.35008578
##
## Goodness-of-fit criteria
##                               1-mle-norm 2-mle-gamma
## Akaike's Information Criterion 1172.264 1163.170
## Bayesian Information Criterion 1177.664 1168.571
```

Gamma distribution has better fit

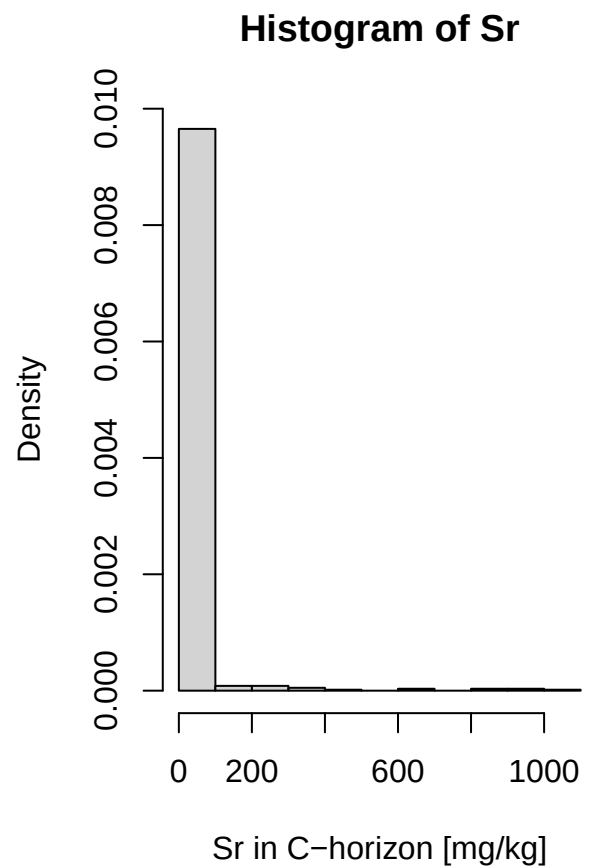
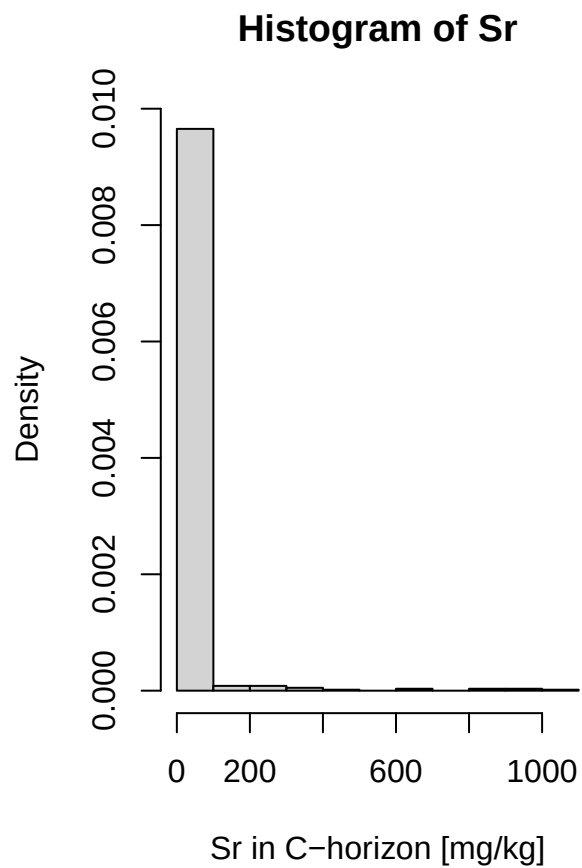
Kola data

```
chorizon<-read.csv("chorizon.csv")
```

```
Be<-chorizon[, "Be"]
Sr<-chorizon[, "Sr"]
plot(Be, Sr,
      xlab="Be in C-horizon [mg/kg]", ylab="Sr in C-horizon [mg/kg]", cex.lab=1.2, pch=3, cex=0.7)
```

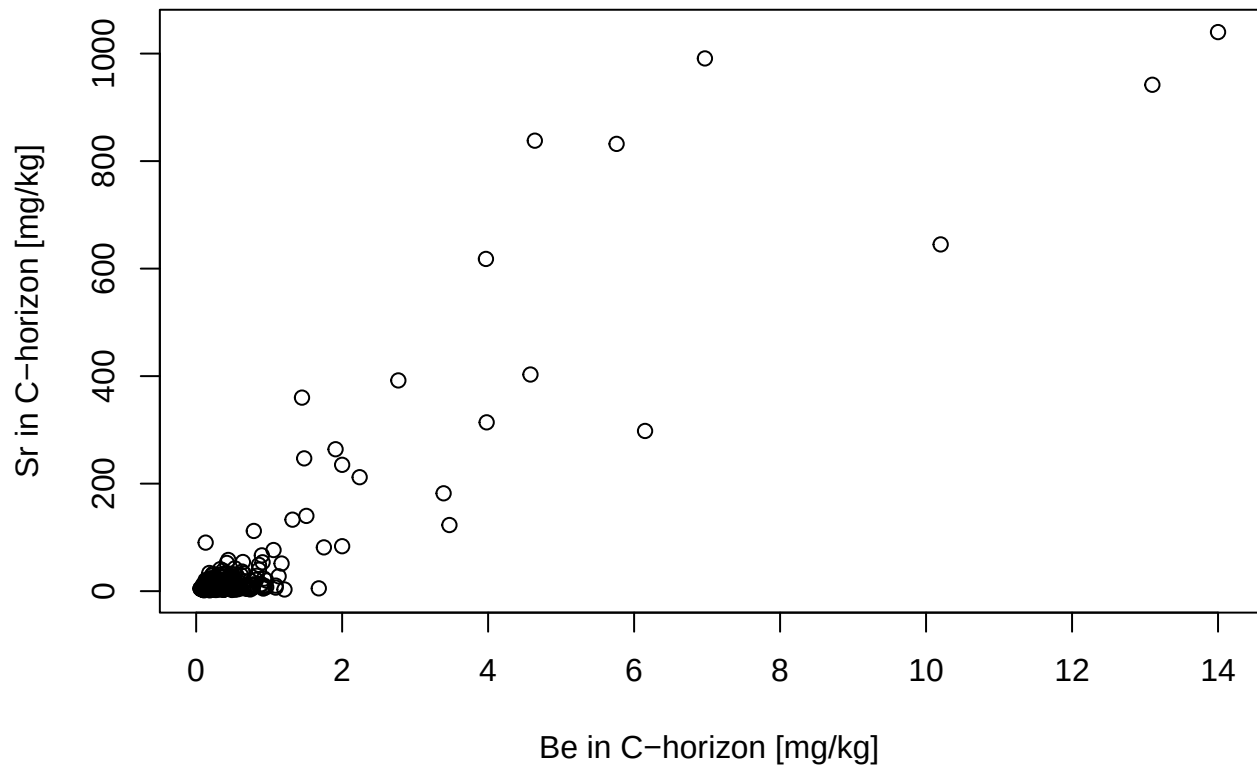


```
par(mfrow=c(1,2))
hist(Sr,xlab="Sr in C-horizon [mg/kg]",freq=FALSE)
hist(Sr,xlab="Sr in C-horizon [mg/kg]",freq=FALSE)
```



```
par(mfrow=c(1,1))
```

```
plot(Be, Sr,xlab="Be in C-horizon [mg/kg]",  
      ylab="Sr in C-horizon [mg/kg]")
```



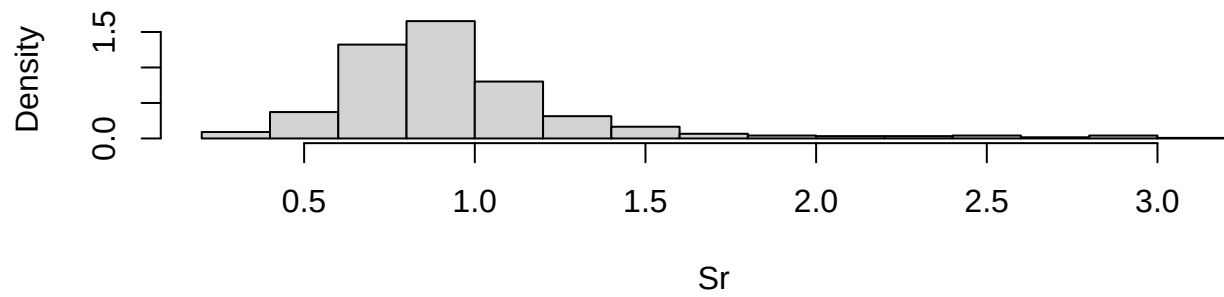
Pearson correlation

```
cor((Be), (Sr))
```

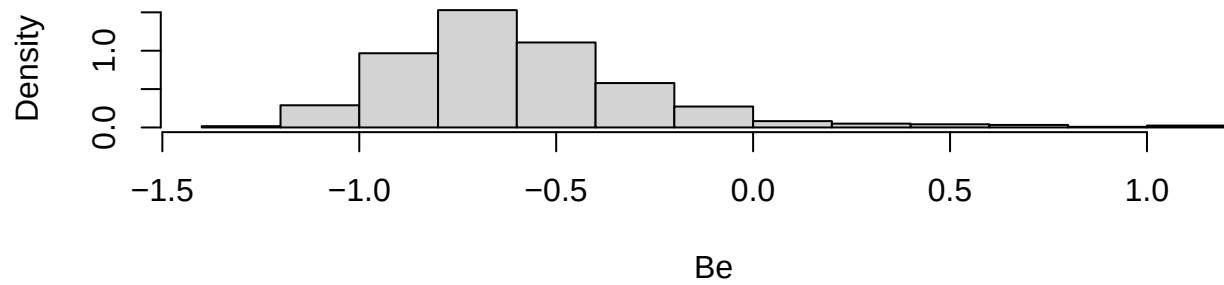
```
## [1] 0.9043162
```

```
par(mfrow=c(2,1))  
hist(log10(Sr),xlab="Sr",freq=FALSE)  
hist(log10(Be),xlab="Be",freq=FALSE)
```

Histogram of $\log_{10}(\text{Sr})$

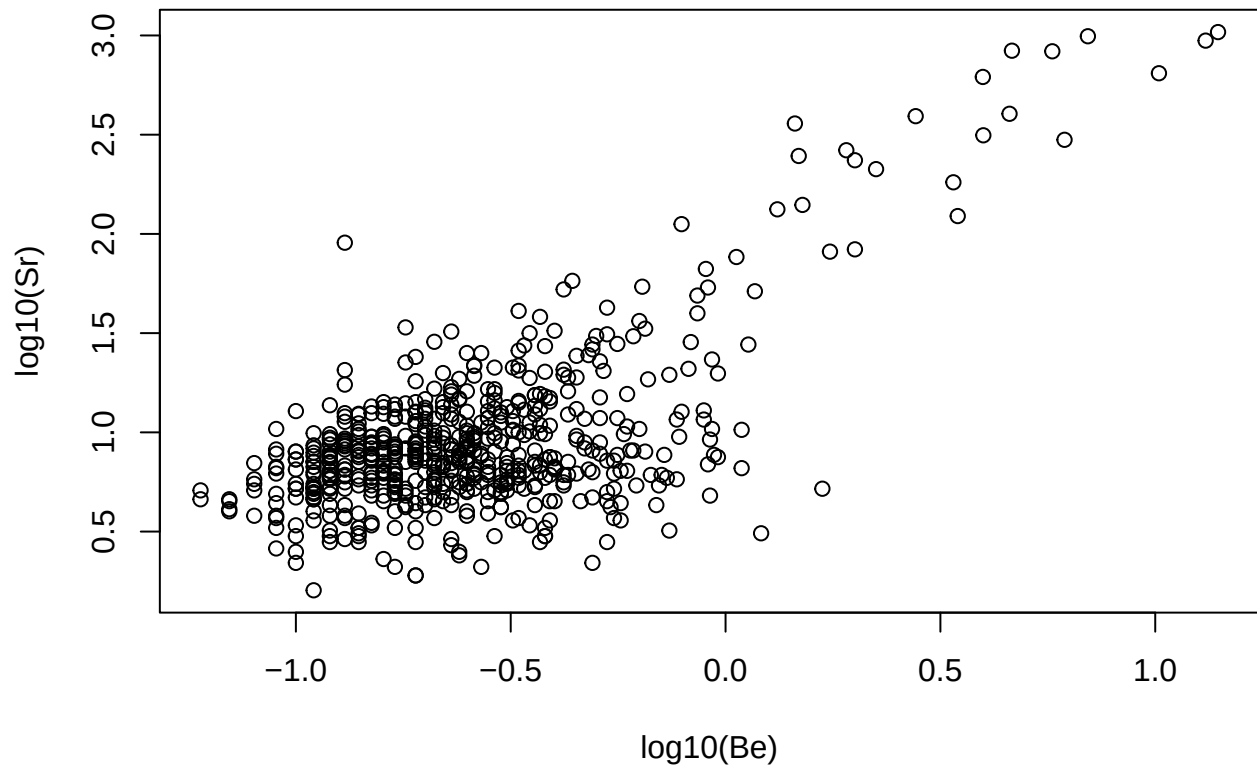


Histogram of $\log_{10}(\text{Be})$



```
par(mfrow=c(1,1))
```

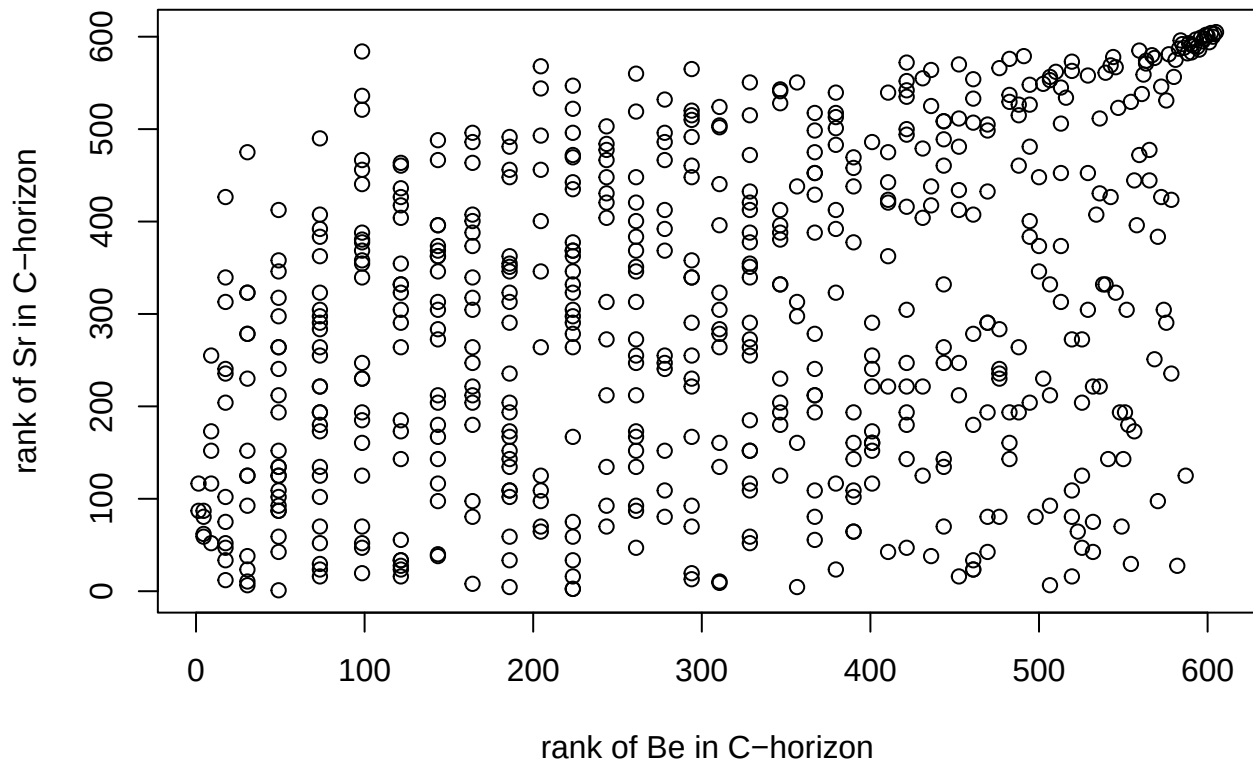
```
plot(log10(Be), log10(Sr))
```



```
cor.test(log10(Be),log10(Sr),
         alternative = "two.sided", method = "pearson")
```

```
##
## Pearson's product-moment correlation
##
## data: log10(Be) and log10(Sr)
## t = 21.667, df = 603, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.6143094 0.7041970
## sample estimates:
##      cor
## 0.6616235
```

```
plot(rank(Be), rank(Sr),
     xlab="rank of Be in C-horizon",ylab="rank of Sr in C-horizon")
```

```
cor((Be), (Sr),method="spearman")

## [1] 0.3670434

cor.test(log10(Be),log10(Sr),
         alternative = "two.sided", method = "spearman")

## Warning in cor.test.default(log10(Be), log10(Sr), alternative = "two.sided", :
## Cannot compute exact p-value with ties

##
## Spearman's rank correlation rho
##
## data: log10(Be) and log10(Sr)
## S = 23360795, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.3670434

cor.test(log10(Be),log10(Sr),
         alternative = "two.sided",method = "kendall")

##
## Kendall's rank correlation tau
##
## data: log10(Be) and log10(Sr)
## z = 9.5745, p-value < 2.2e-16
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
## tau
## 0.2639465
```

```
x<-chorizon[,c("Ca","Cu","Mg","Na","P","Sr","Zn")]
print(round(cor(log10(x),method="spearman"),4))
```

```
##           Ca      Cu      Mg      Na      P      Sr      Zn
## Ca 1.0000 0.2938 0.2624 0.7948 0.6223 0.6977 0.0833
## Cu 0.2938 1.0000 0.7674 0.2619 0.1781 0.3519 0.6599
## Mg 0.2624 0.7674 1.0000 0.1627 0.2122 0.3861 0.7107
## Na 0.7948 0.2619 0.1627 1.0000 0.3839 0.6321 0.0468
## P  0.6223 0.1781 0.2122 0.3839 1.0000 0.4684 0.1985
## Sr 0.6977 0.3519 0.3861 0.6321 0.4684 1.0000 0.4109
## Zn 0.0833 0.6599 0.7107 0.0468 0.1985 0.4109 1.0000
```

Oak data

```
oak<-read.table("oak.txt",header=TRUE)
```

```
Species<-as.factor(oak$Species)
Neighbor<-as.factor(oak$Neighbor)
table(Species,Neighbor)
```

```
##           Neighbor
## Species Black Red White
## Black    27  12   20
## Red      14 104   59
## White    20  62  138
```

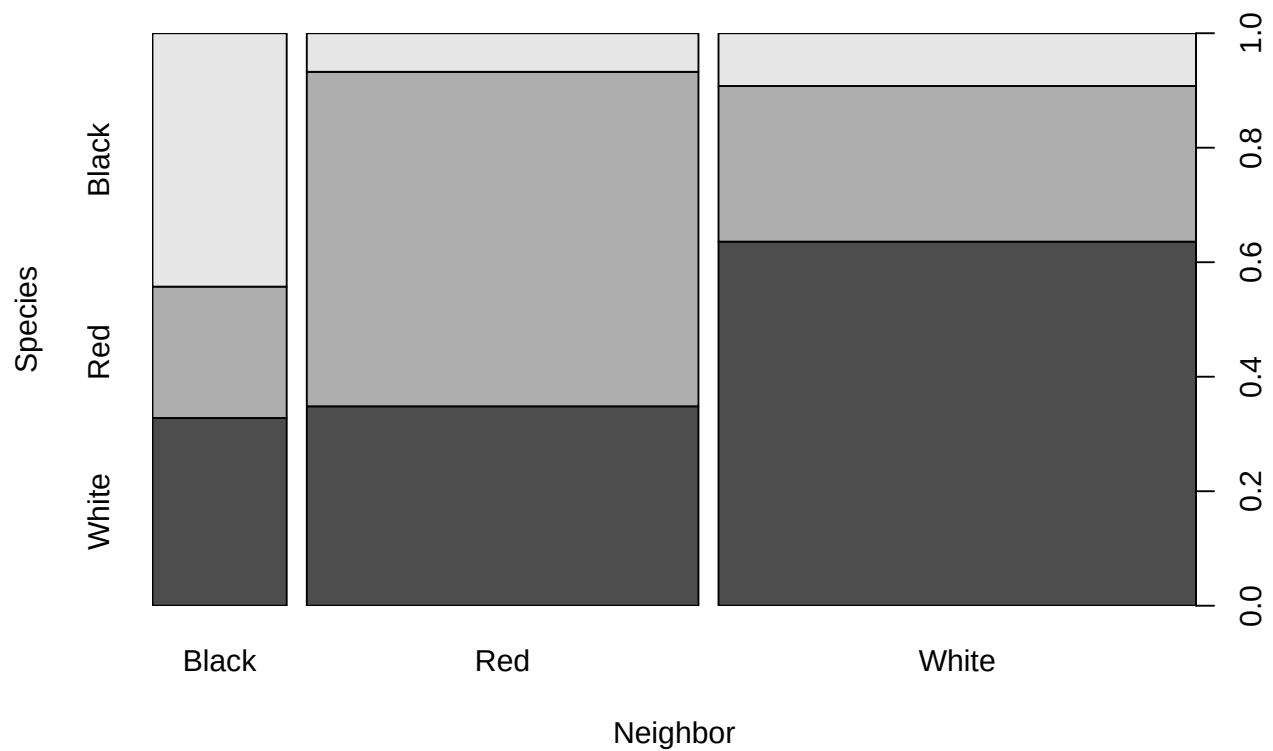
```
proportions(table(Species,Neighbor))
```

```
##           Neighbor
## Species Black Red White
## Black 0.05921053 0.02631579 0.04385965
## Red 0.03070175 0.22807018 0.12938596
## White 0.04385965 0.13596491 0.30263158
```

```
proportions(table(Species,Neighbor),margin=2)
```

```
##           Neighbor
## Species Black Red White
## Black 0.44262295 0.06741573 0.09216590
## Red 0.22950820 0.58426966 0.27188940
## White 0.32786885 0.34831461 0.63594470
```

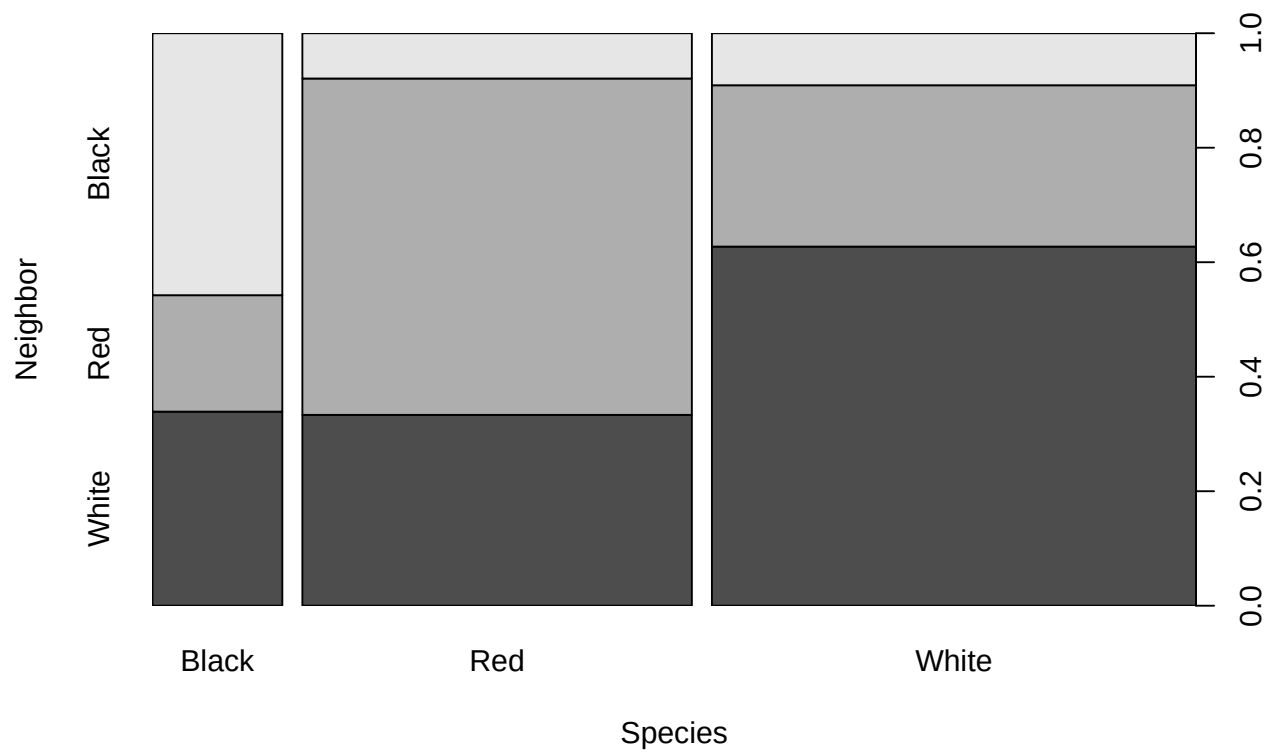
```
plot(Species~Neighbor)
```



```
proportions(table(Species,Neighbor),margin=1)
```

```
##      Neighbor
## Species   Black      Red      White
##   Black 0.45762712 0.20338983 0.33898305
##   Red   0.07909605 0.58757062 0.33333333
##   White 0.09090909 0.28181818 0.62727273
```

```
plot(Neighbor~Species)
```



Test for independence

```
chisq.test(Species,Neighbor)
```

```
##  
## Pearson's Chi-squared test  
##  
## data: Species and Neighbor  
## X-squared = 103.27, df = 4, p-value < 2.2e-16
```